

Verrell's Law

Reduced Formal Draft v1.1

Purpose. This document presents a reduced formal systems draft of Verrell's Law. It is scoped as a general state-evolution framework with memory-weighted bias, not as a completed physics proof.

IP note. The encoding operator G is intentionally left abstract beyond high-level structure. Its concrete decomposition is reserved for a separate CAAI instantiation layer and should be treated as protected implementation logic.

1. Core statement

Verrell's Law: A system's future state is not determined solely by its present configuration and current inputs, but also by retained information from prior states. That retained information acts as a bias field over the set of possible future outcomes, altering which outcomes are more likely to emerge under observation or selection.

future emergence = present conditions + memory-weighted bias

2. State definition

Let the total system state at time t be:

$$X_t = (S_t, M_t, O_t)$$

where S_t is the present system state, M_t is the memory state consisting of retained structured information from prior states, and O_t is the observation state, input, or measurement context. Memory is therefore treated as an active term in system evolution rather than an external record.

3. Candidate future states

Let the system at time t admit a set of candidate future states:

$$Y_t = \{y_1, y_2, \dots, y_n\}$$

These candidate states may represent physical transitions, behavioural outputs, emergent attractors, or other discretised future branches of the system.

4. Memory-biased selection law

The probability of transition into candidate state y_i is given by:

$$P(y_i | S_t, O_t, M_t) = \exp(U(y_i; S_t, O_t) + \lambda B(y_i; M_t)) / \sum \exp(U(y_j; S_t, O_t) + \lambda B(y_j; M_t))$$

Here U is the present-state or observation-fit term, B is the memory-bias term, and $\lambda \geq 0$ is the coupling strength of memory to selection. If $\lambda = 0$, the system is memoryless. If $\lambda > 0$, retained information reshapes the probability landscape and introduces path dependence.

5. Collapse / emergence rule

Once the probability distribution is defined, realised emergence can occur in either of two ways:

Deterministic selection: $y_t = \operatorname{argmax} P(y_i | S_t, O_t, M_t)$

Stochastic collapse: $y_t \sim P(\cdot | S_t, O_t, M_t)$

Verrell's Law therefore permits either winner-take-all selection or probabilistic collapse from a memory-shaped distribution.

6. Structured memory state

In the minimal draft, memory was treated as a single object. In practice, that is often too crude. A more general formulation is:

$$M_t = \{M_t^{(1)}, M_t^{(2)}, \dots, M_t^{(K)}\}$$

Each memory class or region can represent different persistence regimes such as persistent anchors, weighted intermediate memories, or fast-decay ephemeral traces. The bias term then generalises either as a joint function of all classes or in decomposed form:

$$B(y_i; M_t) = \sum w_k B_k(y_i; M_t^{(k)})$$

This allows different memory classes to exert different forms and strengths of influence on future-state selection.

7. Memory update law

If memory biases future emergence, memory itself must evolve. In reduced form:

$$M_{t+1} = (1 - \alpha)M_t + \alpha G(S_t, O_t, y_t)$$

For structured memory systems, the more general form is class-specific:

$$M_{t+1}^{(k)} = (1 - \alpha_k)M_t^{(k)} + \alpha_k G_k(S_t, O_t, y_t)$$

Small α_k implies slow write and high persistence. Large α_k implies rapid update and faster turnover. This lets different memory layers behave differently without breaking the framework.

8. Encoding operator G

The encoding operator is the main open variable in the theory. In abstract form:

$$G : (S_t, O_t, y_t) \rightarrow M_{t+1}$$

It specifies how realised system states become retained structure. In practical systems, G may be gated, structured, and class-dependent. A useful decomposition is:

$$G_k = \Gamma_k(S_t, O_t, y_t, \sigma_t, \nu_t, \rho_t, \pi_t)$$

where σ_t may denote salience, ν_t novelty or surprise, ρ_t reinforcement or recurrence, and π_t governor or policy permission. This does not force a particular implementation. It only states that encoding is not arbitrary and may depend on relevance, novelty, recurrence, and constraint logic.

In concrete implementations, G may perform salience gating, write-strength thresholding, anchor promotion, decay-class assignment, forgetting or demotion, and policy-governed admission. The document intentionally stops at this boundary.

9. Emergence as recursive bias propagation

The full system recursion is therefore:

$$\begin{aligned}
X_t &= (S_t, M_t, O_t) \\
P(y_i | S_t, O_t, M_t) &\propto \exp(U(y_i; S_t, O_t) + \lambda B(y_i; M_t)) \\
y_t &= \text{select from } P(\cdot | S_t, O_t, M_t) \\
M_{t+1}^{(k)} &= (1 - \alpha_k) M_t^{(k)} + \alpha_k G_k(S_t, O_t, y_t) \\
S_{t+1} &= H(S_t, O_t, y_t)
\end{aligned}$$

This defines a memory-weighted, recursively updated state-evolution system.

10. Scope and optional electromagnetic specialisation

This draft claims that systems with retained internal information do not evolve only from present inputs; they evolve through present inputs filtered by memory-weighted bias. That is a general systems claim first. It may later be specialised to cognition, adaptive agents, AI middleware, biological memory systems, or, with additional work, physical field models.

It does not yet claim replacement of quantum mechanics, relativity, or standard field theory. If later specialised to electromagnetic substrates, the memory-bias term may be extended as $B(y_i; M_t, E_t, B_t)$ or equivalent field-coupled forms, but that remains future work rather than the present starting point.

11. Falsifiability conditions

The reduced form fails if any of the following hold:

- Adding M_t and $B(\cdot)$ produces no measurable predictive or explanatory improvement over a memoryless model.
- Systems with different histories but identical present states show indistinguishable future-state distributions.
- The memory term can always be absorbed into U without loss, meaning memory adds no independent structure.
- No stable encoding rule exists such that retained information has repeatable downstream effect.
- A multi-class memory model performs no better than an undifferentiated single-state memory model in systems where layered retention is claimed.

12. Proposition: path-dependent divergence

Consider two systems A and B such that $S_t^A = S_t^B$, $O_t^A = O_t^B$, but $M_t^A \neq M_t^B$. If $B(y_i; M_t^A) \neq B(y_i; M_t^B)$ for at least one candidate y_i , and $\lambda > 0$, then in general the induced future-state distributions differ. Two systems with identical present conditions but different retained histories may therefore exhibit different future-state distributions.

13. Direct relevance to Collapse Aware AI

In CAAI terms, S_t is the current conversational or behavioural state, O_t is the new user input or observation context, M_t is the continuity state composed of weighted moments and anchors, y_i are candidate responses or behaviours, U is immediate relevance or governor-compliant task fit, B is memory-weighted continuity bias, λ is memory influence strength, and G_k is the salience, novelty, recurrence, and governor-controlled write function by memory class.

CAAI is therefore not merely inspired by the law. It is a computational instantiation of it. That gives Verrell's Law a concrete engineering embodiment without pretending that software alone proves cosmology.

14. Tightened summary sentence

Verrell's Law states that retained information from prior system states acts as a bias field over future state selection, causing emergence to be history-dependent rather than determined by present conditions alone.

Publication note: the reduced draft should remain clean and defensible. The concrete composition of the encoding operator G is reserved for later protected CAAI instantiation work rather than this public-facing formal systems note.